ICT283 Assignment 2 - Application Test Plan

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Expected outputs for the correlation were calculated in Excel (=PEARSON); averages and stdev were cross-checked against the data. Fill the 'Passed (Y/N)' column (shaded) by running each test yourself, add notes under Comments, then export to PDF. Do not falsify results. Test 6.1 records a genuine, observed limitation.

No.	Description / Condition Tested	Test Data / Values	Expected Output	Passed (Y/N)	Comments
Option 1 - Average wind speed for a given month and year					
1.1	Valid month & year with data	Option 1; month=6; year=2014	Average speed: 4.7 km/h Sample stdev: 8.4	Y	
1.2	Month/year with no data	Option 1; month=1; year=1850	A 'Enter date within range' message; no crash	Y	
Option 2 - Average temperature for each month of a year					
2.1	Year with data (per-month list)	Option 2; year=2014	Jan (no data), Feb(no data), Mar 22.8 (sd 5.2), Apr 19.3 (sd 5.0), May 16.5 (sd 3.2), Jun 13.2 (sd 4.7), Jul 13.5 (sd 3.7), Aug 15.5, Sep 16.1, Oct 17.8 (sd 4.7), Nov 19.0 (sd 4.6), Dec 21.2 (sd 5.0)	Y	
2.2	Year entirely without data	Option 2; year=1850	A 'Enter date within range' message; no crash	Y	
Option 3 - sPCC correlations for a month across all years					
3.1	Correlation, verified vs Excel =PEARSON	Option 3; month=6	S_T: 0.48 S_R: 0.26 T_R: 0.50 (Excel PEARSON: 0.4779, 0.2650, 0.4956)	Y	
Option 4 - Write wind/temp/solar per month to CSV for a year					
4.1	Valid year writes file with header	Option 4; year=2014	Writes WindTempSolar.csv; header: Month,Wind Speed (stdev, mad),Temperature (stdev, mad),Solar Radiation; e.g. March,20.3(10.9, 8.6),22.8(5.2, 4.1),183.5	Y	
4.2	(stdev, mad) format present	Option 4; year=2014; inspect file	Each average followed by its (stdev, mad) in brackets	Y	
4.3	Year with no data re-prompts	Option 4; year=1850	Reports no data and re-prompts; enter correct year	Y	
4.4	Missing field (blank cell) -> blank between commas	A CSV where one field (e.g. temperature) is blank for a month	Field written blank: e.g. February,4.5(3.1,2.9),,200.3 (two commas)		
Input validation & general					
5.1	Invalid / non-numeric input rejected	Any prompt; e.g. month=13, month=abc, menu=9	Program re-prompts cleanly; no crash and no infinite loop	Y	
5.2	Multi-file load via data_source.txt	Program start; 6 supplied files	Loads 208891 readings across 51 months	Y	
5.3	Columns found by name (any order)	Files with S/T/SR in different column positions	All files load correctly regardless of column order		
5.4	Duplicate date-time rejected	Supplied files (contain repeats)	Repeated date-times ignored; 208891 stored	Y	
5.5	Menu exit	Option 5	Program exits cleanly	Y	
Known limitation					
6.1	File missing an ENTIRE required column	A CSV whose header omits the T (temperature) column completely	Ideally: still load other fields. Actual: whole file is rejected and nothing loads.	N	FAIL (by design). The reader requires all four columns (WAST, S, SR, T) to exist

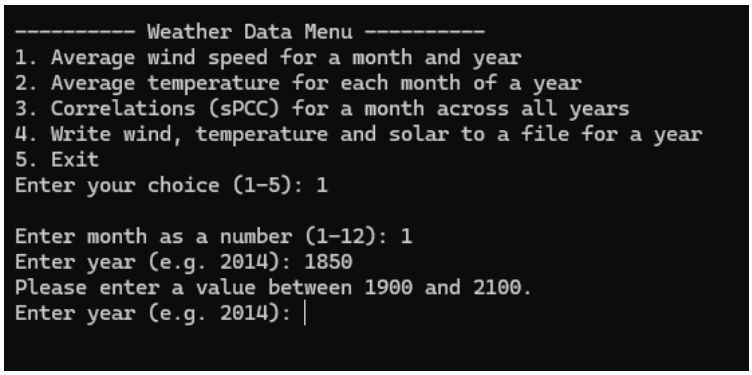
No.	Description / Condition Tested	Test Data / Values	Expected Output	Passed (Y/N)	Comments
					and rejects a file missing one, treating it as invalid input. This differs from a blank CELL in an existing column, which IS handled (test 4.4). Recorded as a known limitation, not a silent bug.

EVIDENCE (SCREENSHOTS):

1.1:



1.2:



2.1:

```
---------- Weather Data Menu ----------
1. Average wind speed for a month and year
2. Average temperature for each month of a year
3. Correlations (sPCC) for a month across all years
4. Write wind, temperature and solar to a file for a year
5. Exit
Enter your choice (1-5): 2

Enter year (e.g. 2014): 2014

2014
January: No Data
February: No Data
March: average: 22.8 degrees C, stdev: 5.2
April: average: 19.3 degrees C, stdev: 5.0
May: average: 16.5 degrees C, stdev: 3.2
June: average: 13.2 degrees C, stdev: 4.7
July: average: 13.5 degrees C, stdev: 3.7
August: average: 15.5 degrees C, stdev: 4.1
September: average: 16.1 degrees C, stdev: 4.0
October: average: 17.8 degrees C, stdev: 4.7
November: average: 19.0 degrees C, stdev: 4.6
December: average: 21.2 degrees C, stdev: 5.0
```

2.2:

```
---------- Weather Data Menu ----------
1. Average wind speed for a month and year
2. Average temperature for each month of a year
3. Correlations (sPCC) for a month across all years
4. Write wind, temperature and solar to a file for a year
5. Exit
Enter your choice (1-5): 2

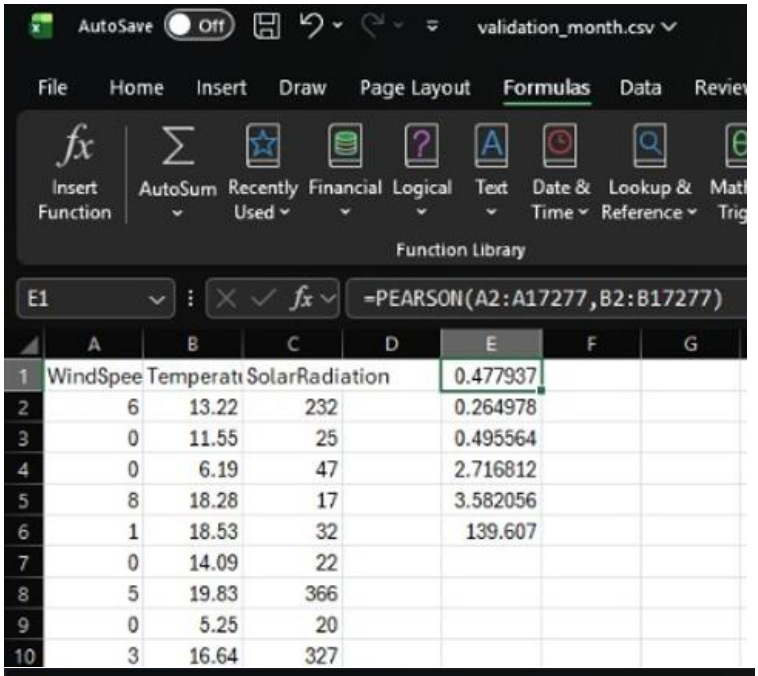
Enter year (e.g. 2014): 1850
Please enter a value between 1900 and 2100.
```

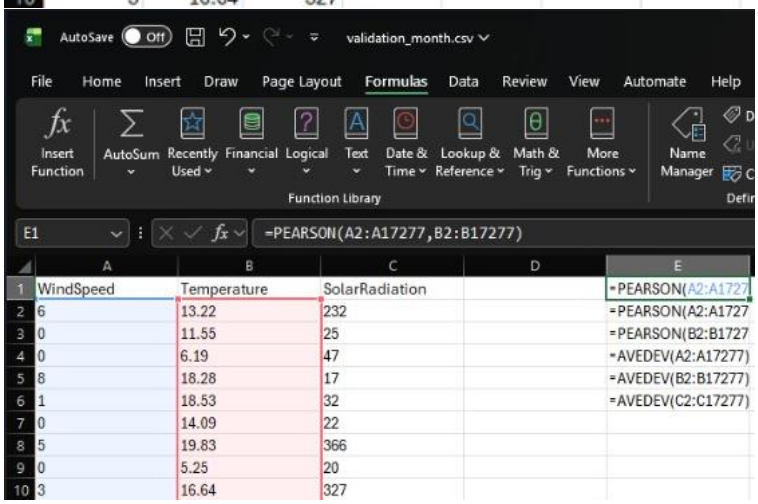
3.1:

```
---------- Weather Data Menu ----------
1. Average wind speed for a month and year
2. Average temperature for each month of a year
3. Correlations (sPCC) for a month across all years
4. Write wind, temperature and solar to a file for a year
5. Exit
Enter your choice (1-5): 3

Enter month as a number (1-12): 6

Sample Pearson Correlation Coefficient for June
S_T: 0.48
S_R: 0.26
T_R: 0.50
```



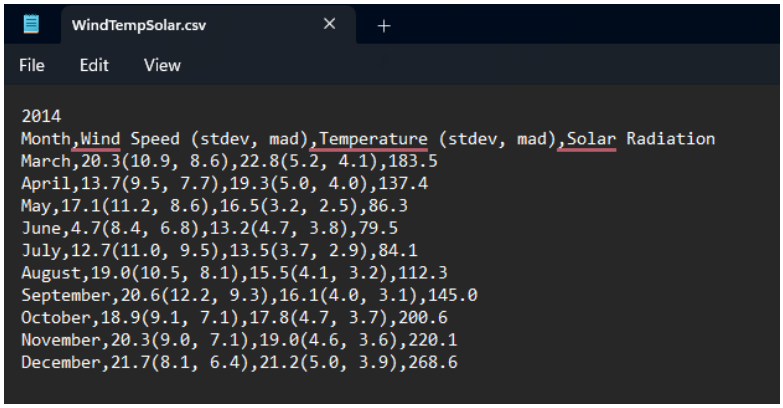


4.1:

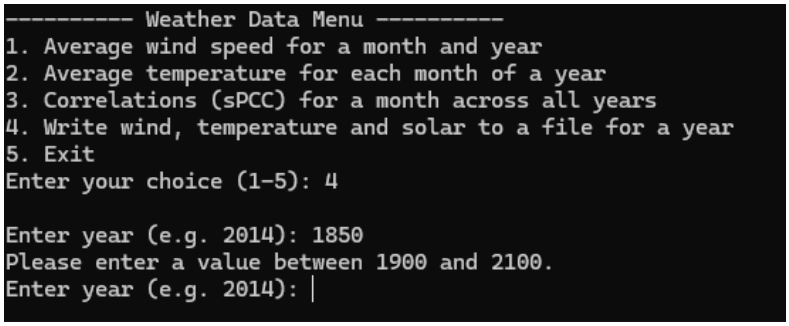
```
---------- Weather Data Menu ----------
1. Average wind speed for a month and year
2. Average temperature for each month of a year
3. Correlations (sPCC) for a month across all years
4. Write wind, temperature and solar to a file for a year
5. Exit
Enter your choice (1-5): 4

Enter year (e.g. 2014): 2014
Data for 2014 written to WindTempSolar.csv
```

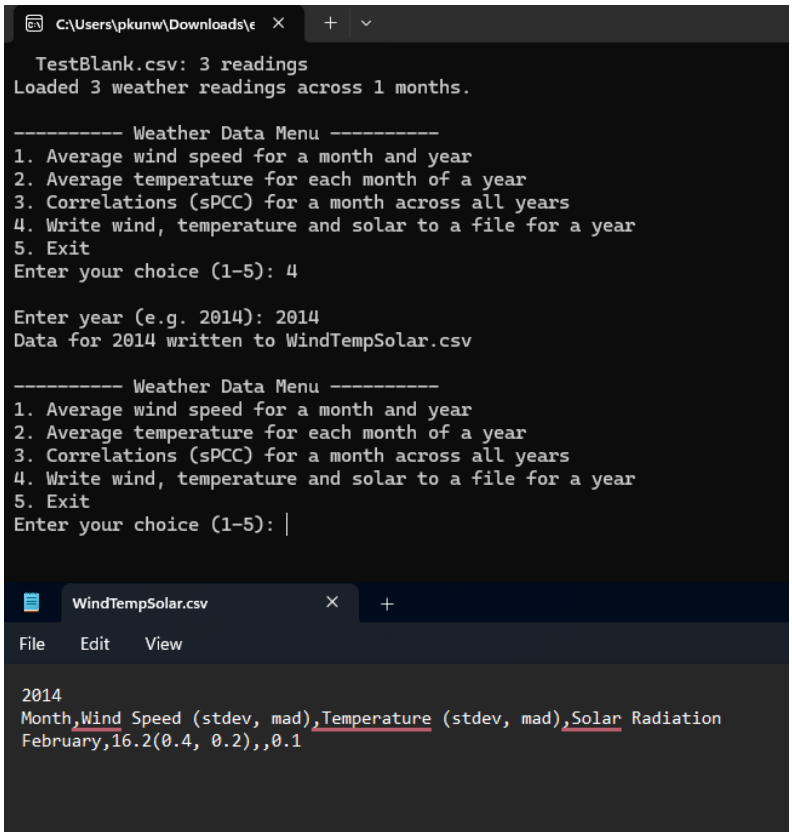
4.2:


WindTempSolar.csv
File
Edit
View
2014
Month,Wind Speed (stdev, mad),Temperature (stdev, mad),Solar Radiation
March,20.3(10.9, 8.6),22.8(5.2, 4.1),183.5
April,13.7(9.5, 7.7),19.3(5.0, 4.0),137.4
May,17.1(11.2, 8.6),16.5(3.2, 2.5),86.3
June,4.7(8.4, 6.8),13.2(4.7, 3.8),79.5
July,12.7(11.0, 9.5),13.5(3.7, 2.9),84.1
August,19.0(10.5, 8.1),15.5(4.1, 3.2),112.3
September,20.6(12.2, 9.3),16.1(4.0, 3.1),145.0
October,18.9(9.1, 7.1),17.8(4.7, 3.7),200.6
November,20.3(9.0, 7.1),19.0(4.6, 3.6),220.1
December,21.7(8.1, 6.4),21.2(5.0, 3.9),268.6

4.3:


---------- Weather Data Menu ----------
1. Average wind speed for a month and year
2. Average temperature for each month of a year
3. Correlations (sPCC) for a month across all years
4. Write wind, temperature and solar to a file for a year
5. Exit
Enter your choice (1-5): 4
Enter year (e.g. 2014): 1850
Please enter a value between 1900 and 2100.
Enter year (e.g. 2014):

4.4:


C:\Users\pkunw\Downloads\e
TestBlank.csv: 3 readings
Loaded 3 weather readings across 1 months.
---------- Weather Data Menu ----------
1. Average wind speed for a month and year
2. Average temperature for each month of a year
3. Correlations (sPCC) for a month across all years
4. Write wind, temperature and solar to a file for a year
5. Exit
Enter your choice (1-5): 4
Enter year (e.g. 2014): 2014
Data for 2014 written to WindTempSolar.csv
---------- Weather Data Menu ----------
1. Average wind speed for a month and year
2. Average temperature for each month of a year
3. Correlations (sPCC) for a month across all years
4. Write wind, temperature and solar to a file for a year
5. Exit
Enter your choice (1-5):
WindTempSolar.csv
File
Edit
View
2014
Month,Wind Speed (stdev, mad),Temperature (stdev, mad),Solar Radiation
February,16.2(0.4, 0.2),,0.1

5.1:

```
---------- Weather Data Menu ----------
1. Average wind speed for a month and year
2. Average temperature for each month of a year
3. Correlations (sPCC) for a month across all years
4. Write wind, temperature and solar to a file for a year
5. Exit
Enter your choice (1-5): 1

Enter month as a number (1-12): 13
Please enter a value between 1 and 12.
Enter month as a number (1-12): abc
Please enter a whole number.
Enter month as a number (1-12): 9
Enter year (e.g. 2014): 2014

September 2014:
Average speed: 20.6 km/h
Sample stdev: 12.2
```

5.2:

```
C:\Users\pkunw\Downloads\I
  MetData_Mar01-2015-Mar01-2016-ALL.csv: 52206 readings
  MetData_Jan01-2010-Jan01-2011-ALL.csv: 52027 readings
  MetData-31-3b.csv: 90 readings
  MetData_Jan01-2011-Jan01-2012-ALL.csv: 51989 readings
  MetData-31-3c.csv: 90 readings
  MetData_Mar01-2014-Mar01-2015-ALL.csv: 52492 readings
Loaded 208891 weather readings across 51 months.

---------- Weather Data Menu ----------
1. Average wind speed for a month and year
2. Average temperature for each month of a year
3. Correlations (sPCC) for a month across all years
4. Write wind, temperature and solar to a file for a year
5. Exit
Enter your choice (1-5): 1
```

5.3:

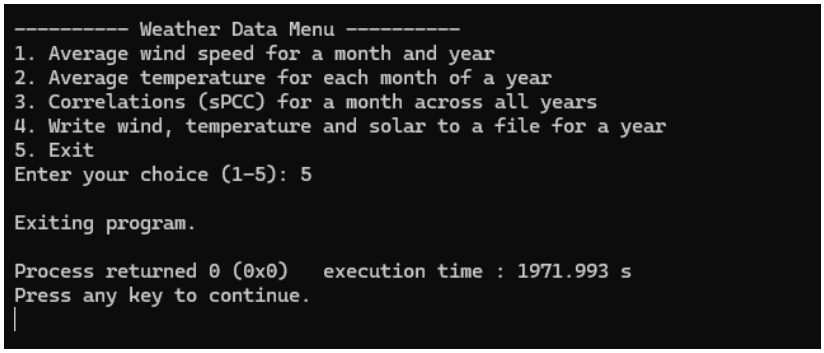
```
Loaded 208891 weather readings across 51 months.
```

NOTE: if the columns weren't being found by name, files with T in a different position would fail to load, and you wouldn't get the full 208,891.

5.4:

```
C:\Users\pkunw\Downloads\I
  MetData_Mar01-2015-Mar01-2016-ALL.csv: 52206 readings
  MetData_Jan01-2010-Jan01-2011-ALL.csv: 52027 readings
  MetData-31-3b.csv: 90 readings
  MetData_Jan01-2011-Jan01-2012-ALL.csv: 51989 readings
  MetData-31-3c.csv: 90 readings
  MetData_Mar01-2014-Mar01-2015-ALL.csv: 52492 readings
Loaded 208891 weather readings across 51 months.
```

5.5



6.1:
The command prompt runs and crashes immediately.